



ME 300: Engineering Thermodynamics

Energy Conservation

Finite time energy conservation

$$\Delta E_{\text{sys}} = \underline{E}_{\text{in}} - \underline{E}_{\text{out}} + \cancel{E_{\text{gen}}}^0 \rightarrow E_{\text{sys}} = KE + PE + U$$

$$\begin{aligned}\underline{d}E_{\text{sys}} &= (\underline{\delta}Q_{\text{in}} + \underline{\delta}W_{\text{in}}) - (\underline{\delta}Q_{\text{out}} + \underline{\delta}W_{\text{out}}) \\ &= (\underline{\delta}Q_{\text{in}} - \underline{\delta}Q_{\text{out}}) + (\underline{\delta}W_{\text{in}} - \underline{\delta}W_{\text{out}}) \\ &= (\underline{\delta}Q_{\text{in}} - \underline{\delta}Q_{\text{out}}) - (\underline{\delta}W_{\text{out}} - \underline{\delta}W_{\text{in}})\end{aligned}$$

$$\begin{aligned}\bar{W}_{\text{out}} &> 0 \quad d\theta > 0 \\ \bar{W}_{\text{in}} &< 0 \quad d\theta < 0\end{aligned}$$

$$dE_{\text{sys}} = \underline{\delta}Q_{\text{in,net}} - \underline{\delta}W_{\text{out,net}}$$

$$\Delta E_{\text{sys}} = \Delta KE + \Delta PE + \Delta U = {}_1Q_2 - {}_1\bar{W}_2$$

First law of thermodynamics

finite time:

$$\Delta E = {}_1Q_2 - {}_1W_2$$

$$\Delta E = \cancel{{}_1Q_2} - {}_1\bar{W}_2$$

$$\Delta E = - \underbrace{{}_1\bar{W}_2}_{<0}$$

$$\Delta E > 0$$

$$\rightarrow \underbrace{{}_1\bar{W}_2}_{<0} = \int_{V_2}^{V_1} P dV > 0, V_2 < V_1$$



In a piston-cylinder system, if the volume decreases adiabatically, does this increase or decrease the energy in the gas?

- a) Increase
- b) Decrease
- c) Stays the same